Thus, the adversary seeks to strike a balance: they aim to detect or destroy hidden messages without making alterations so severe that they provoke widespread dissent. This motivates the formalization of tampering functions and defines the scope of robustness required for steganographic systems.

4.2 Adversary Capabilities

We assume a man-in-the-middle adversary with the following capabilities:

- **Intercept and Analyze**: The adversary can intercept and analyze all communication between the sender and receiver.
- **Modify Stegotext**: The adversary can modify the stegotext, either by introducing small structural changes or by generating semantically equivalent paraphrases.
- **Linguistic Coherence**: The adversary has the ability to produce plausible modifications using automated or manual natural language processing techniques.

Despite these capabilities, the adversary is constrained by the need to avoid provoking widespread dissent among the affected population. This leads to two adversarial models: the **weak threat model**, which assumes limited tampering capabilities, and the **strong threat model**, which assumes more sophisticated attacks.

4.3 Weak Threat Model: Locally Consistent Tampering

This model corresponds to *weak robustness* or *local robustness*, which captures adversaries that make small, localized edits—such as reordering n-grams, substituting synonyms, or inserting short phrases. These changes typically preserve surface fluency and often preserve meaning, but this is not always the case. For example, inserting the word "not" may leave most of the structure unchanged while inverting the semantic intent of the sentence.

In the weak threat model, the adversary's modifications are limited to small structural changes to the message. These changes are modeled by a family of tampering functions $\mathcal{F}$, specifically (k, ϵ)-locally consistent functions.

Definition of Locally Consistent Tampering. A tampering function $f \in \mathcal{F}_{k,\epsilon}$ ensures that for every input string $x \in \Sigma^*$, the tampered string $f(x)$ agrees with x on at least an ϵ-fraction of its k-length continuous substrings. Formally:

$$\mathcal{F}_{k,\epsilon} = \left\{ f : \forall x \in \Sigma^*, \sum_{i=1}^{|x|-k+1} \xi\left(x[i : i+k-1], f(x)\right) \geq \epsilon \cdot \left(|x| - k + 1\right) \right\},$$

where $\xi(x[i : i+k-1], f(x)) = 1$ if $x[i : i+k-1]$ is a continuous substring of $f(x)$, and 0 otherwise.

Examples of locally consistent tampering include:

- Reordering or shuffling n-grams.
- Deleting or repeating small parts of the message.
- Making minor edits that leave the message's intent intact.

The weak model reflects adversarial actions designed to destroy hidden communication while avoiding overt disruption that might provoke dissatisfaction.

4.4 Strong Threat Model: Semantically Preserving Paraphrasing

This model corresponds to *strong robustness* or *semantic robustness*, which captures adversaries that make broader, meaning-preserving rewrites—such as paraphrasing, stylistic reformulations, or structural transformations. These changes may significantly reshape how the sentence is written while leaving its meaning intact.

The strong threat model assumes a more sophisticated adversary capable of generating paraphrases of the stegotext. These paraphrases:

- **Preserve Semantic Meaning**: The adversary rewrites the text to retain the original meaning, ensuring it remains plausible to the recipient.
- **Maintain Embedding Proximity**: The paraphrase must remain close to the original text in the embedding space, reflecting semantic equivalence.

Definition of Embedding Proximity. Let $E(x)$ denote the embedding of a text $x \in \Sigma^*$.

We define a distance function

$$\text{distance} : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}_{\geq 0},$$

mapping pairs of embeddings in $\mathbb{R}^d$ to nonnegative real values. Any well-defined distance function (e.g., Euclidean distance or cosine similarity) may be used, and will correspond to a particular Locality Sensitive Hash (LSH) function.

A paraphrase x' satisfies:

$$\text{distance}(E(x), E(x')) \leq \delta,$$

where $\delta > 0$ is a small threshold representing allowable semantic deviation.

Examples of adversarial actions in this model include:

- Rewriting sentences using synonyms or alternative phrasing.
- Using NLP tools to generate semantically equivalent variations of the text.
- Making changes to sentence structure while retaining overall meaning.

This strong model reflects adversaries capable of leveraging advanced linguistic and computational techniques to detect or disrupt hidden messages.

4.5 Summary and Comparison of Threat Models

- **Weak Model (Local Robustness)**: The adversary modifies stegotext using (k, ϵ)-locally consistent tampering functions, introducing small structural changes to the text.
- **Strong Model (Semantic Robustness)**: The adversary generates semantically equivalent paraphrases constrained by proximity in the embedding space, representing a more sophisticated attack.

These threat models highlight the balance adversaries seek to strike: they aim to disrupt hidden communication without sufficiently disrupting the message's meaning to provoke retaliation. The weak model aligns with scenarios involving simple structural tampering, while the strong model reflects adversaries equipped with advanced paraphrasing capabilities. Although these definitions are formally incomparable—neither strictly implies the other—they differ in how well they align with the motivations of real-world adversaries. For instance, inserting the word "not" may satisfy

the weak definition by preserving local structure, but it changes the meaning of the message and would likely be avoided by an adversary under the motivating conditions for this work. In this sense, the strong (semantic) definition more accurately captures the kinds of transformations we aim to defend against. We retain the terms *weak* and *strong* throughout the paper to match our formal definitions and maintain continuity with prior work. At the same time, we use *local* and *semantic* as alternative names to clarify the nature of the edits each model targets. Notably, the weak model connects naturally to existing paradigms like watermarking, which can be seen as a special case of steganography and has historically focused on robustness to small, localized changes. Our strong definition generalizes this by targeting meaning-preserving transformations, motivating a new class of robust stegosystems.

5 Defining Robust Steganography

In this section, we formally define the notion of a robust steganographic scheme. While standard security definitions ensure that hidden communication is indistinguishable from ordinary covertext, they fail to address scenarios where adversaries actively modify stegotext messages. As described in the threat model, adversaries aim to detect or disrupt hidden communication without provoking societal backlash, constraining their tampering to realistic limits.

To address these practical considerations, we introduce robustness definitions that guarantee successful decoding under two levels of adversarial tampering:

- **Weak robustness**, which accounts for small structural changes modeled by locally consistent tampering functions.
- **Strong robustness**, which accounts for semantically equivalent paraphrasing constrained by proximity in the embedding space.

These definitions provide formal guarantees against adversarial modifications that align with the weak and strong threat models. We first recall the standard definition of a symmetric-key steganographic scheme, and then define robustness under both the weak and strong threat models introduced earlier.

5.1 Symmetric-key Steganography

In this work, we focus on the symmetric-key setting. In this setting, the sender and receiver share a secret key k, and communicate over a *covertext channel* Ch. This channel is characterized by an alphabet Σ and a mapping $\mathcal{D}$ that maps a history $\boldsymbol{h} = (h_1, \ldots, h_\ell)$ to a probability distribution over Σ^*.

Following the definitions in [15], a symmetric-key steganographic scheme is parameterized by a *covertext channel* with alphabet $\mathcal{T}$ and distribution $\mathcal{D}$, and consists of a tuple $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$, where:

- $\mathsf{KeyGen}(1^\lambda) \to k$: A randomized key generation algorithm that takes the security parameter λ and outputs a symmetric key k.
- $\mathsf{Enc}(k, m, \boldsymbol{h}) \to st$: A randomized encoding algorithm that takes a key k, a message m, and a history $\boldsymbol{h} = (h_1, \ldots, h_\ell)$ of previously sent stegotext messages, and outputs a stegotext message st.